An Evaluatable Retail Customer Service Agent

A bounded agent and the framework that evaluates its decisions

LangGraph Langfuse

Final project proposal

Problem and audience

The situation

Customers say “I want to return this item” or “My promo code doesn’t work” without giving everything needed.

The decision

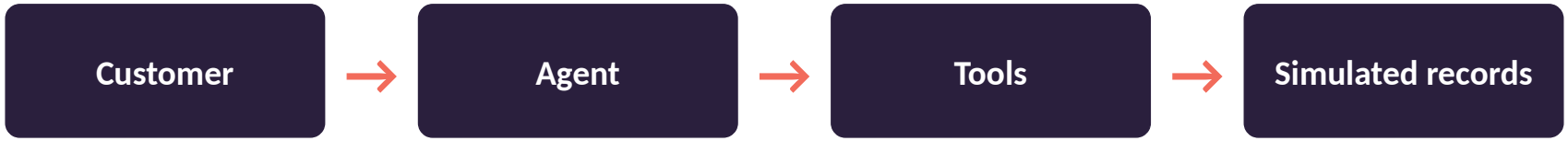
A representative must choose what to ask, which records to check, and what policy allows.

Who it is for

Retail customer service teams face this problem. For us, it is a learning project on fake data; the skills carry into future apps.

Can an agent choose the right tool, gather enough evidence, and follow policy while the conversation unfolds?

Proposed solution



types a request

asks, decides, follows policy

functions the agent calls to check records

fake accounts, orders, products (SQLite)

What a user can do

- Ask for a return (size, color, or stain issues)
- Report an invalid or expired promo code
- Request a cancellation, upgrade, or downgrade

Only these three request families are supported.

What we deliver

- Interactive customer service demo
- Reproducible test cases
- Reviewed reference labels
- Evaluation report on where it fails

Approach

Data

ABCD supplies three things:

- Scenario facts fill a simulated SQLite company database
- Transcripts become the conversation history
- Agent guidelines supply the policy rules

Test cases

Checkpoints pause a conversation at a decision point and ask the agent for its next move.

- Tool-use: before a recorded action. Which tool, which arguments?
- Policy-decision: after new information. Ask, respond, or act?

Workflow

- Read the conversation so far
- Decide: ask, query records, check policy, or respond
- Repeat until resolved

LangGraph only; design in progress.

Model: GPT-5.4 mini (planned)

Tracing and evaluation: Langfuse records each run

Demo: a light web app on fake data, for learning, not a product

Scope and risks

In scope

- Three request families
- A bounded agent on simulated data
- 128 reviewed checkpoints (dev and test)
- Demo within a $20 budget

Out of scope

- Training a new model
- Real accounts or persistent orders

Main risks

We have no labels yet

ABCD records what the original representative did, not what is correct. Different LLMs will draft labels for each test case, and we review them, starting where they disagree.

One decision at a time

Checkpoint tests check a single decision, not a whole conversation. If time allows, we also run a few complete conversations.

Plan and success criteria

Four-week plan

Week 1	Tools on SQLite records; load policy guidelines; select checkpoints
Week 2	LangGraph workflow with tracing; draft and review labels
Week 3	Policy integration, first evaluation runs, error analysis
Week 4	Final evaluation, demo, and report

What the demo could look like

Customer: My promo code doesn't work.
Agent: What's the code and your order number?
Customer: SAVE20, order 1234.
Agent: That code expired last week, so I can't apply it.

Trace: ask → look up order → check policy → reply

We will show

- All three request families, end to end
- How often decisions match reviewed labels
- Agent failure types, like acting too early